Machine Learning

Richa Singh

Google classroom code: spjv64lu

Slides are prepared from several information sources on the web and books

About the instructor

- Richa Singh:
  - Professor - CSE, IIT Jodhpur
  - Faculty, IIIT Delhi (2009 – 2019)
  - Fellow, INAE, IEEE, IAPR, NASI and AAIA
  - Distinguished Member, ACM
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  - Associate Program Co-chair, AAAI 2027
  - Associate Editor-in-Chief, Pattern Recognition Journal
  - Founding Co-Editor-in-Chief, ACM AI Letters
  - Program Co-Chair, CVPR 2022

Course Objectives

- To understand various key paradigms for machine learning approaches
- To familiarize with the mathematical and statistical techniques used in machine learning.
- To understand and differentiate among various machine learning techniques.

Learning Outcomes

- The students are expected to have the ability to:
  - To formulate a machine learning problem
  - Select an appropriate pattern analysis tool for analyzing data in a given feature space.
  - Apply pattern recognition and machine learning techniques such as classification and feature selection to practical applications and detect patterns in the data.

Course Content

Fractal I: Supervised Learning

- Introduction: Definitions, Datasets for Machine Learning, Different Paradigms of Machine Learning, Data Normalization, Hypothesis Evaluation, VC-Dimensions and Distribution, Bias-Variance Tradeoff, Regression (Linear) (7 Lectures)
- Bayes Decision Theory: Bayes decision rule, Minimum error rate classification, Normal density and discriminant functions (5 Lectures)
- Parameter Estimation: Maximum Likelihood and Bayesian Parameter Estimation (2 Lectures)

Fractal II: Unsupervised Learning

- Discriminative Methods: Distance-based methods, Linear Discriminant Functions, Decision Tree, Random Decision Forest and Boosting (6 Lectures)
- Feature Selection and Dimensionality Reduction: PCA, LDA, ICA, SFFS, SBFS (4 Lectures)
- Clustering: k-means clustering, Gaussian Mixture Modeling, EM-algorithm (4 Lectures)

Fractal III: Kernels and Neural Networks

- Kernel Machines: Kernel Tricks, SVMs (primal and dual forms), K-SVR, K-PCA (6 Lectures)
- Artificial Neural Networks: MLP, Backprop, and RBF-Net (4 Lectures)
- Foundations of Deep Learning: DNN, CNN, Autoencoders (4 lectures)

Reading Resources

- Textbooks:
  - Pattern Classification, 2nd Edition, Richard O. Duda, Peter E. Hart, David G. Stork, Wiley

- Reference Books:
  - Tom Mitchell, Machine Learning
  - C. Bishop, Pattern Recognition and Machine Learning, Springer
  - K. Murphy, Machine Learning: a Probabilistic Perspective, MIT Press
  - Shalev-Shwartz,S., Ben-David,S., (2014), Understanding Machine Learning: From Theory to Algorithms, Cambridge University Press

Evaluation Components

- Grading
  - Assignments (labs, programming and written): 25%
  - Exams: 50%
  - Project: 15%
  - Quiz: 10%
- Project team size: 2-3 students
- Assignments: individually

Collaboration Policy

- Discussion with friends and colleagues is good… but
  - the objective should be to improve understanding and learning
  - Not getting answers
- If you have discussed with anyone, you should acknowledge who helped you – from the class or outside the class

Plagiarism Policy

- Cheating in assignments/quizzes/projects/
  - First offence: Zero in the evaluation component
  - Second offense: Grade reduction/F grade
- Cheating in exam: F grade
- Misbehavior: Institute guidelines

Any questions regarding administrative guidelines?

What is Machine Learning?

Machine Learning

- What do we understand by learning?
  - Learning is any process by which a system improves performance from experience.”

    – Herbert Simon (1950)
- Machine Learning is the study of algorithms that
  - improve their performance P
  - at some task T
  - with experience E.

  A well-defined learning task is given by <P, T, E>.

- Tom Mitchell (1998)

How to design a learning system

- Understand the problem statement
- Choose exactly what is to be learned
- Choose how to represent the target function
- Choose a learning algorithm to infer the target function from the experience

Based on slide by Ray Mooney

Applications of ML



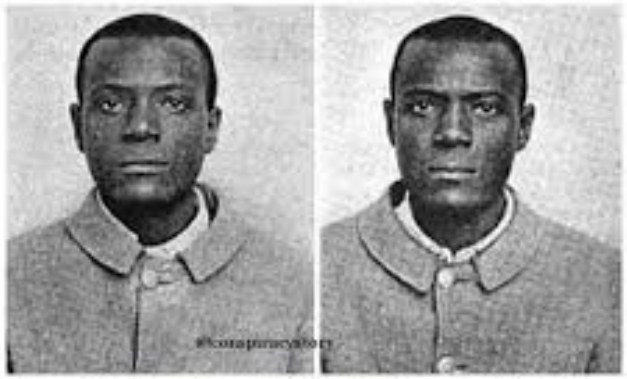
What are the facial expressions?



Weather forecast



What are these letters?



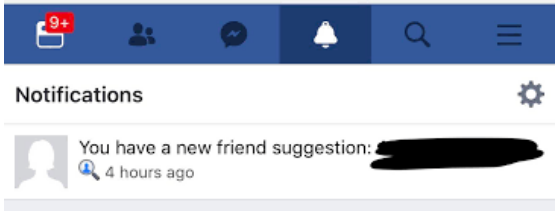
Are they same or different?

Applications of ML







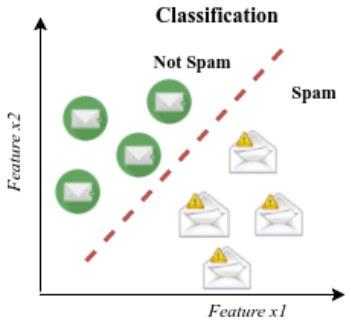


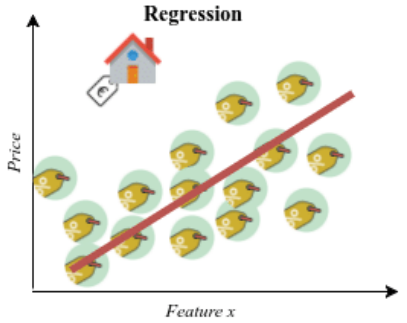




Task, T

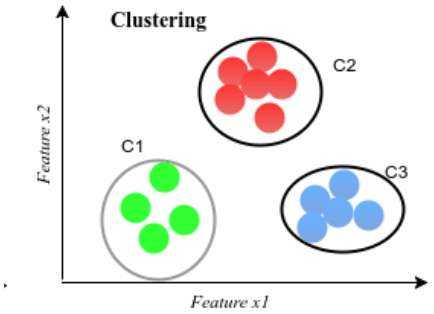
- Classification
- Regression
- Ranking
- Recommendation
- Clustering
- Density estimation
- ….











Performance, P

- Metric used to evaluate the performance of T
  - Classification: error rate or accuracy
  - Regression: mean squared error
  - Density estimation: probability assigned to samples

ML in Practice



- Understand domain, prior knowledge, and goals
- Data integration, selection, cleaning, pre-processing, etc.
- Learn models
- Interpret results
- Consolidate and deploy discovered knowledge

Based on a slide by Pedro Domingos

Machine Learning Pipeline

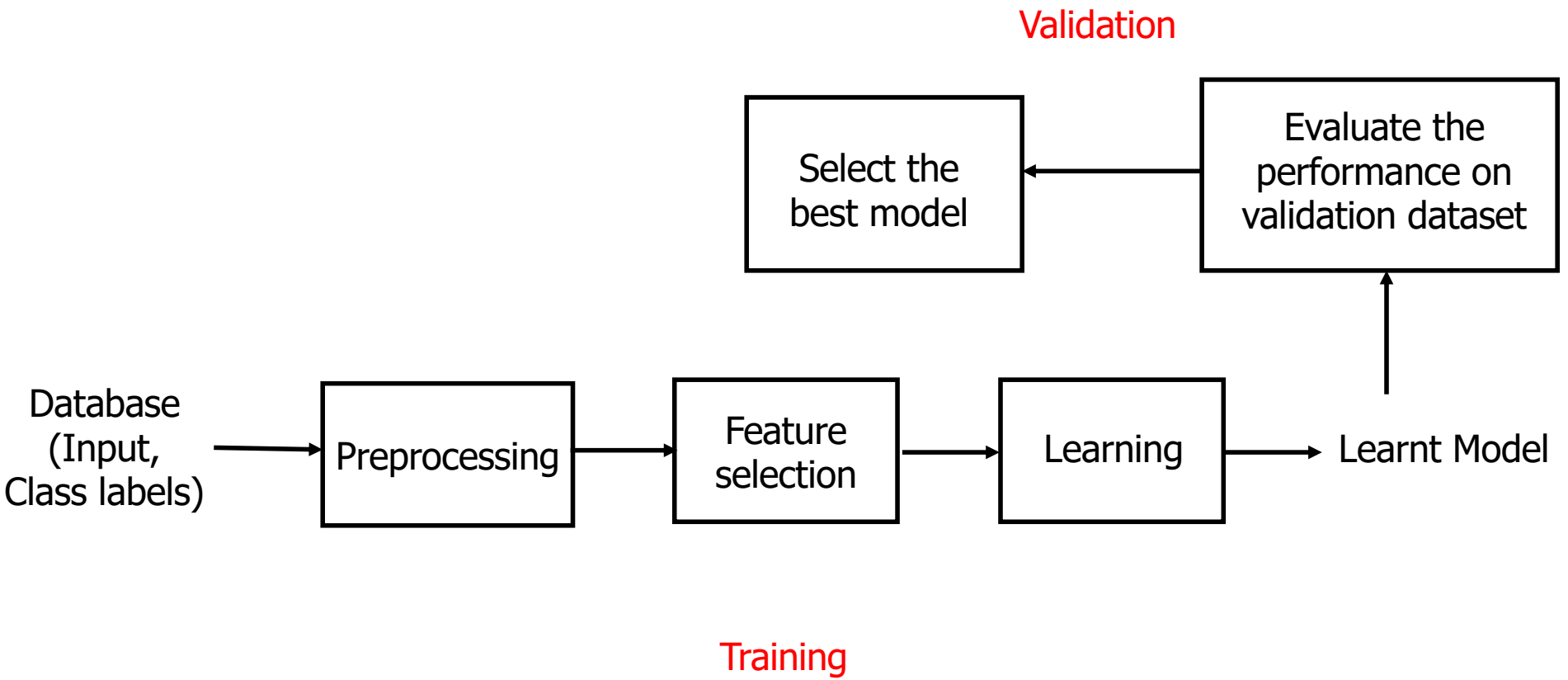
- Three steps:
  - Training
  - Validation
  - Testing

Machine Learning Pipeline



Training

Machine Learning Pipeline



Machine Learning Pipeline

